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Ultrathin Broadband Reflective Optical Limiter

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Optical limiters are nonlinear devices that feature decreasing transmittance with increasing incident optical intensity, and thus can protect sensitive components from high-intensity illumination. The ideal optical limiter reflects rather than absorbs light in its active ("limiting") state, minimizing risk of damage to the limiter itself. Previous efforts to realize reflective (rather than absorbing) limiters were based on embedding nonlinear layers into relatively thick multilayer photonic structures, resulting in substantial fabrication complexity, reduced speed and, in some instances, limited working bandwidth. In this paper, these tradeoffs are overcome by using the insulator-to-metal transition (IMT) in vanadium dioxide (VO_2) to achieve intensity-dependent modulation of resonant transmission through aperture antennas. Due to the large change of optical properties across the IMT, low-quality-factor resonators are sufficient to achieve high on–off ratios in the transmittance of the limiter. As a result, our ultrathin reflective limiter (thickness ≈1/100 of the free-space wavelength) is broadband in terms of operating wavelength (>2 μm at 10 μm) and angle of incidence (up to ≈50° away from the normal). Our analysis of the experimental results via opto-thermal simulations provides insight into limiter performance and is a useful guidance for further engineering efforts.

1. Introduction

The growing ubiquity of high-power light sources[1,2] and the increasing sensitivity of photodetectors and cameras[3,4] motivate designs of optical limiters that can protect delicate optical components and human eyes against high-intensity illumination. Optical limiters are designed to be transparent for low-intensity incident light (open state), but should have gradually decreasing transmittance for increasing intensity (limiting state) such that the output intensity remains roughly constant.[5–9]

Existing optical limiters are primarily based on nonlinear absorption,[9,10] with alternatives including nonlinear refraction,[11] nonlinear scattering,[12,13] or nonlinear reflection.[14] However, each of these mechanisms has drawbacks. The use of nonlinear refraction, scattering, or reflection requires additional care to redirect the refracted or scattered light away from any light-sensitive devices, so nonlinear absorption is typically preferred. A variety of nonlinear absorption mechanisms across many material systems have been used, including reverse saturable absorption in organic/organometallic polymers,[8,15] multi-photon absorption in liquid crystals[16] and graphene and its derivatives,[17] free-carrier absorption in semiconductors,[18] nonlinear absorption in correlated materials,[19] and many others.[6,9,20,21] Despite their popularity, absorptive limiters are often not ideal because absorption of intense incident light can damage the limiter itself, for example, via melting, sputtering, delamination, or ablation.

The ideal solution is thus a limiter that transitions from a high-transmittance state at low intensity to a reflecting state at high intensity. Such devices have recently been proposed and demonstrated using multilayer photonic structures with the nonlinear medium comprising gallium arsenide (GaAs)[22] or germanium–antimony–tellurium (GST).[23] In the first demonstration based on GaAs,[22] the spectral and angular bandwidths were small due to the large quality factor of the photonic-crystal resonator, which was needed to enhance the optical nonlinearity. In the recent demonstration based on GST,[23] a complex multilayer structure with multiple embedded GST layers was used to achieve broadband performance, enabled by the large effective nonlinearity due to the phase change of GST. However, the use of multiple active layers within a thick structure can limit the switching speed, and the nonvolatile phase change in GST means that a limiter based

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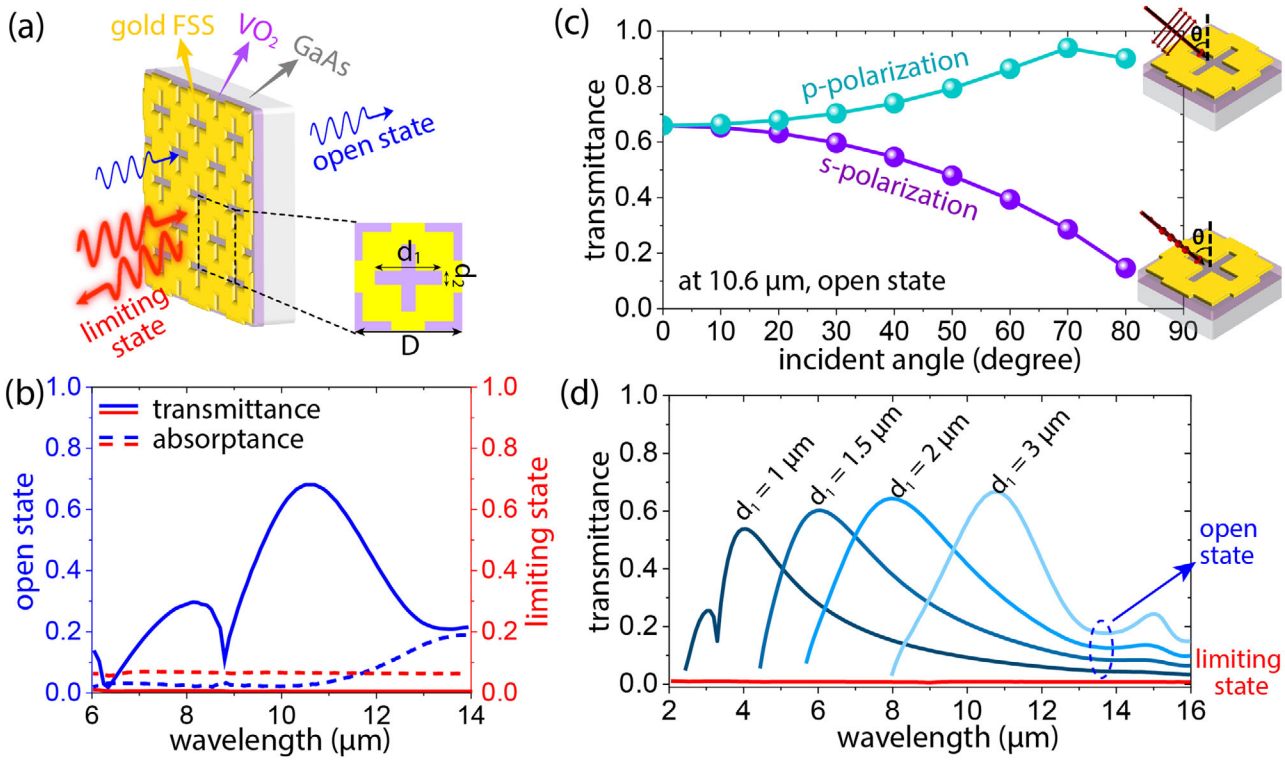

Figure 1. a) Schematic of an optical limiter based on a gold frequency-selective surface (FSS) on top of a thin film of VO_2, with GaAs (001) as the transparent substrate. b) Simulated open- and limiting-state transmittance and absorptance of the structure in "(a)" with $d_1 = 0.2$ μm, $d_2 = 3.1$ μm, and $D = 3.5$ μm. The thickness of gold and VO_2 are 50 nm and 100 nm, respectively. The central wavelength of the open-state transmittance peak is $\lambda = 10.6$ μm, with peak transmittance of 0.7. The limiting-state transmittance is smaller than 0.01, accompanied by absorptance of ≈0.06 at all wavelengths. c) Simulated angle-dependent open-state transmittance of the design in "b" for both s- and p-polarizations at $\lambda = 10.6$ μm. d) The central wavelength of the open-state resonant transmittance is tunable between 4 and 11 μm by changing d_1 from 1 to 3 μm. The value of d_2 was fixed at 0.2 μm. The value of D is always set to $(d_1 + 0.4)$ μm.

on this material can only switch to the limiting state a single time before it must be manually reset.

In this paper, we demonstrate an ultrathin, volatile, reflective optical limiter that only comprises two subwavelength-thickness functional layers and operates over a broad range of wavelengths and incidence angles. This is achieved using resonant transmission through a metallic frequency-selective surface (FSS) comprising low-quality-factor aperture antennas, made optically responsive using the insulator-to-metal transition (IMT) of vanadium dioxide (VO_2), which can be triggered optically via a photothermal process.[24,25] The very large change in optical properties across the IMT[26] enables the strong modulation of resonant transmission through the aperture antennas, resulting in a limiter with broadband transmission in the open state and low transmission and absorption in the limiting state.

2. Design and Simulation

Our design comprises a gold FSS, a thin film of VO_2, and a transparent substrate (GaAs), as shown in **Figure 1**a. The IMT in VO_2 can be thermally triggered by heating to ≈70 °C,[27,28] and can be a source of photothermal nonlinearity[26,29] that is orders of magnitude stronger than conventional nonlinearities (e.g., the Kerr effect[30,31]). Indeed, the limiting characteristics of VO_2 films based on this IMT are well known.[25,32,33] However, metal-phase VO_2 is not nearly as reflective as noble metals (e.g., gold and silver),[26,34] and therefore a nontrivial amount of incident power can still be absorbed by the metal-phase VO_2.

Our use of an FSS on top of a VO_2 film significantly reduces the limiting-state absorption, enabling the use of the IMT in VO_2 for reflective limiting. The gold FSS in our design is an array of close-packed cross-slit aperture antennas within a thin film of gold (Figure 1a) that has high resonant transmittance when the nearby VO_2 film is in the insulating phase. Once the IMT is triggered by sufficiently intense incident light (e.g., intensity of ≈1–100 kW·cm^{-2}, as shown in the next section), the resonant transmittance is expected to decrease quickly via two mechanisms. First, the resonance is frequency shifted by the large change in the real part of the refractive index of VO_2 ($\frac{\Delta n}{n} > 100\%$ in the mid infrared[26]). Second, the increasing loss in the VO_2 further suppresses the amplitude of the resonant transmission. These two mechanisms result in a large reduction of the device transmittance as the VO_2 undergoes the IMT. For the substrate, we chose undoped GaAs (001) since it is mostly transparent across the mid infrared.[35] We used finite-difference time-domain (FDTD) simulations (implemented in Lumerical FDTD) for design and optimization.

The structure was optimized for maximum transmittance at a target wavelength when in the open state, and maximum reflectance when in the limiting state. Specifically, we

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considered three figures of merit: a) high open-state transmittance that ensures high transmission efficiency of low-intensity light; b) low limiting-state transmittance that ensures effective limiting of high-intensity light; and c) low limiting-state absorptance that enables a high damage threshold. For this paper, we chose $\lambda = 10.6$ µm as the central wavelength of the open-state resonant transmittance. Using the temperature-dependent refractive indices of VO_2 that were characterized in our previous work (ref. [26]), we optimized the structural parameters of the aperture antennas, including length (d_1) and width (d_2) of the aperture antenna and the periodicity (D). Note that there are tradeoffs between our three figures of merit. For example, a larger d_2 or a smaller D helps to improve the open-state transmittance but also increases the limiting-state absorptance (see detailed discussion in Supporting Information S1).

After considering these tradeoffs during optimization, we settled on $d_1 = 3.1$ µm, $d_2 = 0.2$ µm, and $D = 3.5$ µm. The thicknesses of gold or VO_2 have much less influence on our figures of merit because they are much smaller than the wavelength (thickness of ≈100 nm ≪ operational wavelength of 10.6 µm). Therefore, we chose the thicknesses of gold and VO_2 to be 50 and 100 nm, respectively. The resulting structure features a broad transmittance band (FWHM > 2 µm) centered at $\lambda = 10.6$ µm in the open-state and close-to-zero transmittance (≈0.008) in the limiting state (Figure 1b). The limiting-state absorptance (≈0.06 at all wavelengths) is significantly reduced compared to that of the bare VO_2 film with no FSS (≈0.2, Supporting Information S2, and more details in *Analysis and Discussion*), thus enhancing the damage threshold of the device. The limiting-state absorptance can be further decreased by reducing the area density of the aperture antennas; however, this comes at the cost of the amplitude and bandwidth of the open-state transmittance peak (Supporting Information S1). Due to the subwavelength thickness of the FSS-VO_2 layer and the symmetry of the aperture antennas, the open-state transmittance remains high for oblique incidence up to ≈50° from the normal for both s- and p-polarizations (Figure 1c). Because insulating thin-film VO_2 has low optical loss in the infrared from ≈2 to 11 µm,[26] the transmittance peak of the limiter can be shifted to anywhere within this window by changing the length of aperture antennas (Figure 1d).

3. Experiments

Our fabrication process includes four steps: VO_2 deposition, e-beam lithography, gold evaporation, and lift-off (**Figure 2**a). First, VO_2 was magnetron sputtered onto double-side-polished undoped GaAs (001) wafers (see details in *Experimental Section*). The thickness of the resulting film is (105 ± 6) nm, measured by scanning electron microscopy (SEM) imaging of the cross-section and atomic force microscopy (AFM) imaging of the surface. The stoichiometry of the film was confirmed by Raman spectroscopy measurements at 30 and 100 °C (see details in Supporting Information S3). The IMT of the film was confirmed by temperature-dependent near-normal-incidence transmission/reflection measurements using a Fourier-transform spectrometer (FTS) (Figure 2b). All spectra were collected at temperatures between 30 and 100 °C (first heating, then cooling) with steps of 2 °C and a ramping rate of 1 °C min^{-1}. The as-grown film featured an IMT from ≈70 to 82 °C when heated (Figure S3c, Supporting Information).

The as-grown VO_2 film features broadband transmittance of ≈0.6 in the insulating phase and relatively low transmittance when thermally biased to the metallic phase (Figure 2c). The features at ≈10 µm of the metal-phase VO_2 are possibly due to a very thin surface oxide of V_2O_5 on the as-grown film[26,36] (more details in Supplementary Information S3). We note that, as shown later, such thin surface oxide does not substantially affect the performance of our fabricated limiter. The absorptance spectrum at 100 °C (absorptance = 1 − transmittance − reflectance) indicates that the metal-phase VO_2 has a relatively high absorptance of ≈0.3 at all wavelengths.

Then, we built a 50-nm gold FSS on top of the as-grown VO_2 via steps as shown in Figure 2a (see details in *Experimental Section*). At 30 °C (device in the open state), the fabricated FSS-VO_2 limiter features a peak transmittance of 0.45 at $\lambda = 9.8$ µm, while the transmittance decreased to ≈0.03 when the temperature was increased to 100 °C (device in the limiting state) (Figure 2d). The measured reflectance in the limiting state is >0.9 at all wavelengths (Figure 2f), indicating that the limiting-state absorptance is <0.06 (Figure 2d), which is substantially reduced compared to that of the bare VO_2 film (Figure 2c). To demonstrate that our design is feasible for wavelengths across the mid infrared, we also fabricated samples with different aperture lengths. The measurements (Figure 2g) showed that our limiters with different aperture lengths (d_1 = 1, 1.5, 2, 2.5, and 3 µm) have central wavelengths of resonant transmittance ranging from 4 to 11 µm, as expected based on the simulations in Figure 1d. We attribute the transmittance decrease for $d_1 = 3$ µm to the increasing optical loss in VO_2 at ≈11 µm.[26]

The measured spectra of the fabricated FSS-VO_2 limiter (Figure 2d) essentially agree with our simulations (Figure 1b), though there are discrepancies in the amplitudes and peak positions of the open-state transmittance. The amplitude of experimental open-state transmittance is lower than that of the simulation, likely because we did not consider the backside of the substrate in the FDTD simulation. After taking it into account, the simulated peak value matches well with that of the experimental result (Supporting Information S4). In practice, one can manufacture anti-reflection coatings on the backside of the GaAs substrate to achieve the performance similar to simulations. Also, the open-state central wavelength of the fabricated limiter is blue-shifted by ≈0.8 µm compared to our simulation results primarily due to two experimental errors. First, the aperture length and width in the fabricated FSS are ≈3 and ≈0.17 µm, respectively (Supporting Information S4), both smaller than those we used in simulations. Second, the refractive indices of VO_2 that we used for simulation were not extracted from the film that we used for fabrication. Small differences in refractive-index values are expected between VO_2 films, though they share similar tendencies in this wavelength region.[26]

Despite the shift of the transmittance peak, our fabricated limiter still features reasonable open-state transmittance of 0.36 at $\lambda = 10.6$ µm, enabling us to use a continuous-wave CO_2 laser to test our design. In our intensity-dependent transmission measurement setup (**Figure 3**a), the sample was mounted on a heat stage with a through-hole aperture used to thermally bias the device, which was necessary since the maximum incident

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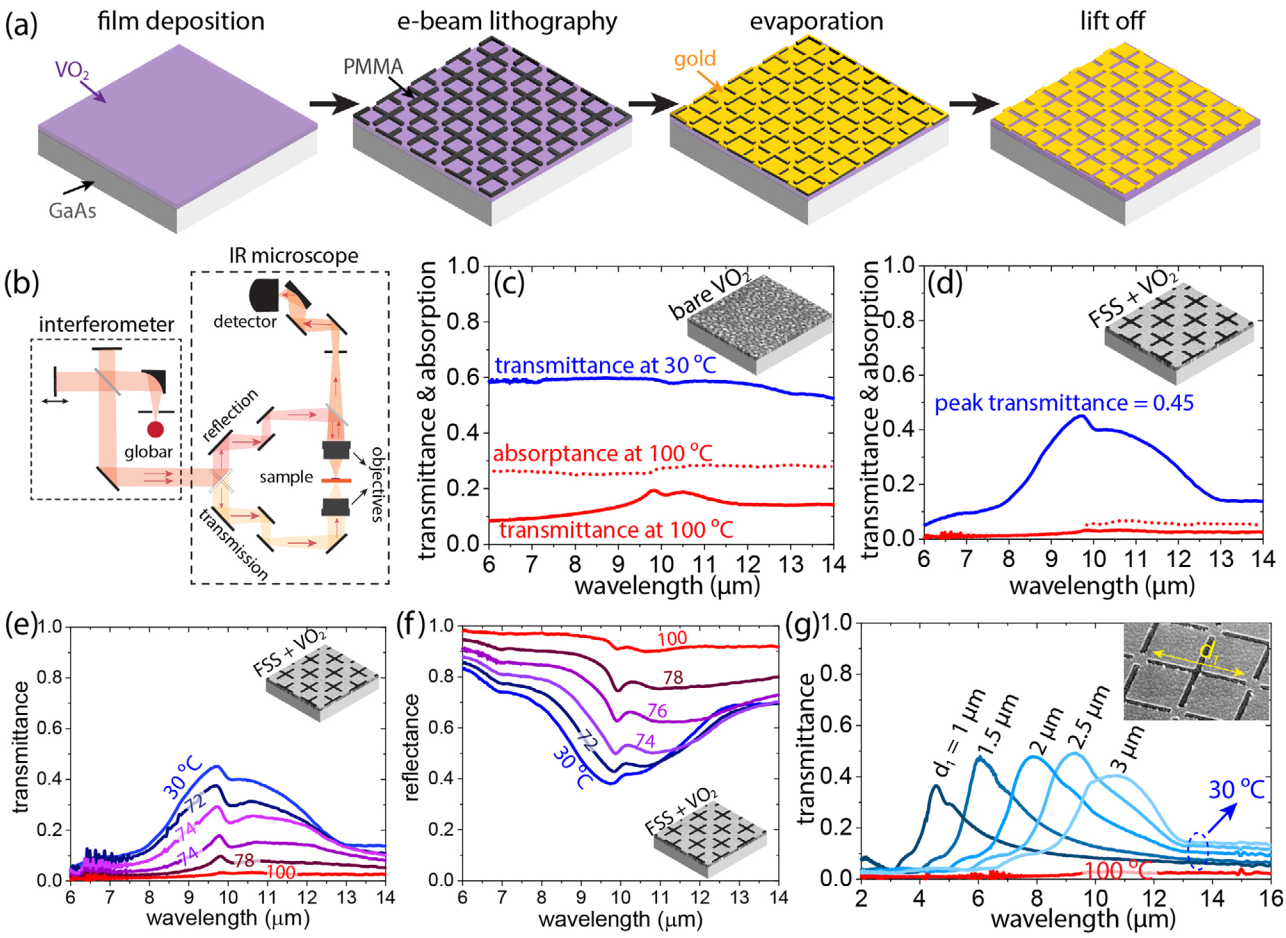

Figure 2. a) Fabrication flow: we first synthesized ≈100-nm VO_2 on GaAs wafer using magnetron sputtering. The as-grown film went through an e-beam lithography and development process, resulting in an array of cross-shape blocks of PMMA. Then, we evaporated 50 nm of gold on top, and performed a lift-off process, leaving an array of cross-slit apertures within the gold film. b) A schematic of simplified optical path of our transmission and reflection measurements, using an infrared microscope (Bruker Hyperion 2000) attached to a Fourier-transform spectrometer (Bruker Vertex 70). c) Transmittance and absorptance spectra of the as-grown VO_2 film, for both insulating (30 °C) and metallic (100 °C) phases. d) Transmittance and absorptance spectra of the FSS-VO_2 limiter at 30 °C (open state) and 100 °C (limiting state). e,f) Transmittance and reflectance measurements of our fabricated FSS-VO_2 limiter when VO_2 in its pure insulating phase (30 °C), pure metallic phase (100 °C), and intermediate phases across the IMT (at 72, 74, 76, and 78 °C). g) Measured transmittance of the FSS-VO_2 limiters with aperture lengths d_1 of 1, 1.5, 2, 2.5, and 3 μm, for both open (30 °C) and limiting (100 °C) states. The top surfaces of the inset figures in (c–g) are SEM images of the corresponding fabricated samples.

intensity achieved in our setup was not high enough to photothermally trigger the IMT from room temperature (see more details in *Analysis and Discussion*). The linearly polarized laser beam was first expanded using an infrared concave lens, then collimated (beam diameter ≈10 mm) and focused onto the sample using two infrared convex lenses (focal length $f = 10$ cm), resulting in near-normal incidence (NA ≈0.05) with a maximum incident intensity of ≈6.5 kW·cm^{-2} (Supporting Information S5). A rotating wire-grid infrared polarizer was used as a tunable attenuator, yielding an incident power range from less than 0.1 to ≈190 mW. Two more convex lenses were used to re-focus light that passed through the sample onto a power meter (Thorlabs S132C). To avoid the thermal hysteresis of the VO_2[37] from influencing the measurements, we heated the sample from room temperature to a set bias temperature for each data point, then turned on the laser beam and recorded the transmitted power once the bias temperature was reached.

We measured the transmission of both the bare VO_2 film and fabricated FSS-VO_2 limiter for a series of incident powers and at different temperatures (i.e., temperature of the stage). For the bare VO_2 film (Figure 3b), we did not observe nonlinear transmission (i.e., limiting behavior) when the temperature was set at or below 70 °C, which is at the edge of the IMT for increasing temperature (Figure S3c, Supporting Information). Limiting behavior became apparent when the film was thermally biased to 74 °C, which is already an intermediate temperature within the IMT. The throughput power was limited to ≈55 mW at high-power incidence (i.e., incident power greater than 120 mW). In contrast, the FSS-VO_2 limiter featured obvious nonlinear transmission at ≈70 °C for incident power starting at 30 mW (the pink curve in Figure 3c). The transmitted power was limited to ≈25 mW for incident power from 90 to 190 mW. Furthermore, at a lower biasing temperature of 62 °C, which is far outside the IMT temperature range, we also observed limiting behavior when the incident

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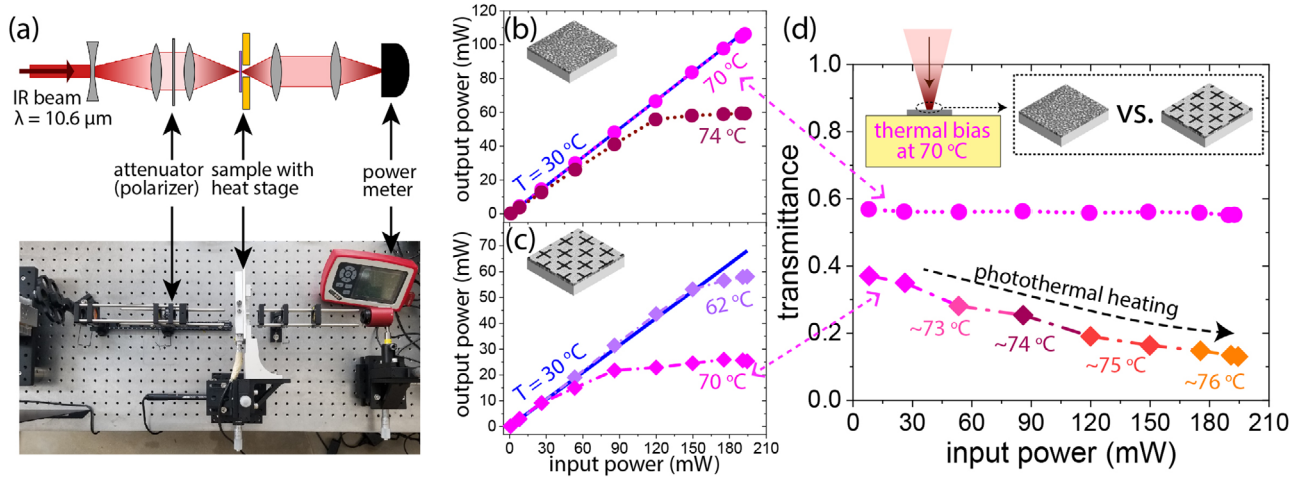

Figure 3. a) Our measurement setup using a continuous-wave CO_2 laser ($\lambda = 10.6$ μm) as the light source and a heat stage for thermal biasing. b) Power-dependent transmission measurements of the bare VO_2 film when it was thermally biased at $T = 30$, 70, and 74 °C. c) Power-dependent transmission measurements of the FSS-VO_2 limiter when it was biased at $T = 30$, 62, and 70 °C. d) A comparison between the power-dependent transmittance of the bare VO_2 film and the FSS-VO_2 limiter, when the heat stage was set at 70 °C. Each transmittance can be correlated with the local temperature at the beam spot using our previous temperature-dependent transmittance measurements, that is, results in Figures 2e and 4b.

power was increased above 150 mW. Such limiting behavior was not found in the bare VO_2 film using the same experimental conditions.

4. Analysis and Discussion

Our intensity-dependent transmission measurements (Figure 3b,c) can be qualitatively understood using our previous FDTD simulation (Figure 1) and temperature-dependent measurements using FTS (Figure 2c-f and **Figure 4**a,b). As expected from the initial simulations (Figure 1b), the introduction of the FSS resulted in much lower limiting-state absorption and higher transmittance contrast between the two states compared to the bare VO_2 film. Somewhat counterintuitively, the introduction of the FSS also resulted in limiting action starting at lower incident powers or lower bias temperature (i.e., a reduction of the limiting threshold). As shown in Figure 3d, when biased at 70 °C, the bare VO_2 film featured constant (i.e., linear) transmittance at all input powers, while the FSS-VO_2 limiter featured a decreasing (i.e., nonlinear) transmittance as the incident power increases, indicating that the limiter was photothermally heated to intermediate temperatures in the IMT region at the beam spot. For each transmittance, we can approximate the corresponding local temperature within the VO_2 film at the beam spot using our temperature-dependent transmittance measurements shown in Figure 2e.

The limiting threshold and required thermal bias depend strongly on the open-state absorptance of the device. In the case of bare thin-film VO_2, the open-state absorptance is very low (≈0.02, Figure 4a), so a high incident power is required to trigger the IMT. Alternatively, the film can be thermally biased part of the way into the IMT (e.g., at 74 °C, the dark red curve in Figure 3b), where the absorptance has begun to rise even in the absence of incident light. Due to the absorptance of the nanostructured gold film, the introduction of the FSS results in much-higher open-state absorptance of ≈0.12 in our device (Figure 4a), enabling limiting action that begins far outside the IMT (e.g., at 62 °C, the purple curve in Figure 3c). Furthermore, in the FSS-VO_2 limiter, both the higher open-state absorptance and the increase of absorptance at the onset of the IMT (Figure 4a) result in faster turn-on compared to the bare VO_2 film. As the IMT evolves toward the pure metallic phase, the subsequent decrease of the absorptance starts to slow down the IMT, resulting in limiting performance for a broad intensity range (Figure 3c).

To quantitatively understand our intensity-dependent measurements, we built a photothermal simulation using COMSOL Multiphysics. Our model consists of an axisymmetric geometry of five domains centered on the center axis of the laser beam (Figure 4c). The heat generation within the device is due to laser power absorbed in the gold FSS and VO_2 layers (domains 1 and 3 in Figure 4c). The laser beam used in our model had a uniform intensity distribution within a radius of 50 μm. We used a continuous gold film to represent the gold FSS in our thermal modeling, which is expected to be a valid assumption because the area density of gold in the FSS is high (≈0.88) and the apertures are small. We assumed the thermal conductivity and specific heat capacity of VO_2 to be 6 W $(m \cdot K)^{-1}$ and 690 J $(K \cdot kg)^{-1}$, respectively.[38,39] Note that these values correspond to VO_2 in its insulating phase, but their change due to the IMT is not dramatic (less than 20% across the IMT),[39] so for simplicity we set them to be constant. Thermal properties of gold and GaAs were taken from refs. [40, 41].

Our model works as follows to calculate the transmittance given a particular incident intensity for a given bias temperature. With no laser irradiation, the device is in thermal equilibrium at the given bias temperature. The transient thermal simulation is initiated by heat flux induced into the device by laser irradiation. For each increasement in time, the simulation returns a transient temperature distribution within the device, which is used to update the absorptance based on the measured temperature-dependent absorption in Figure 4a. The resulting absorptance is then fed into the simulation at the next time step. This coupled

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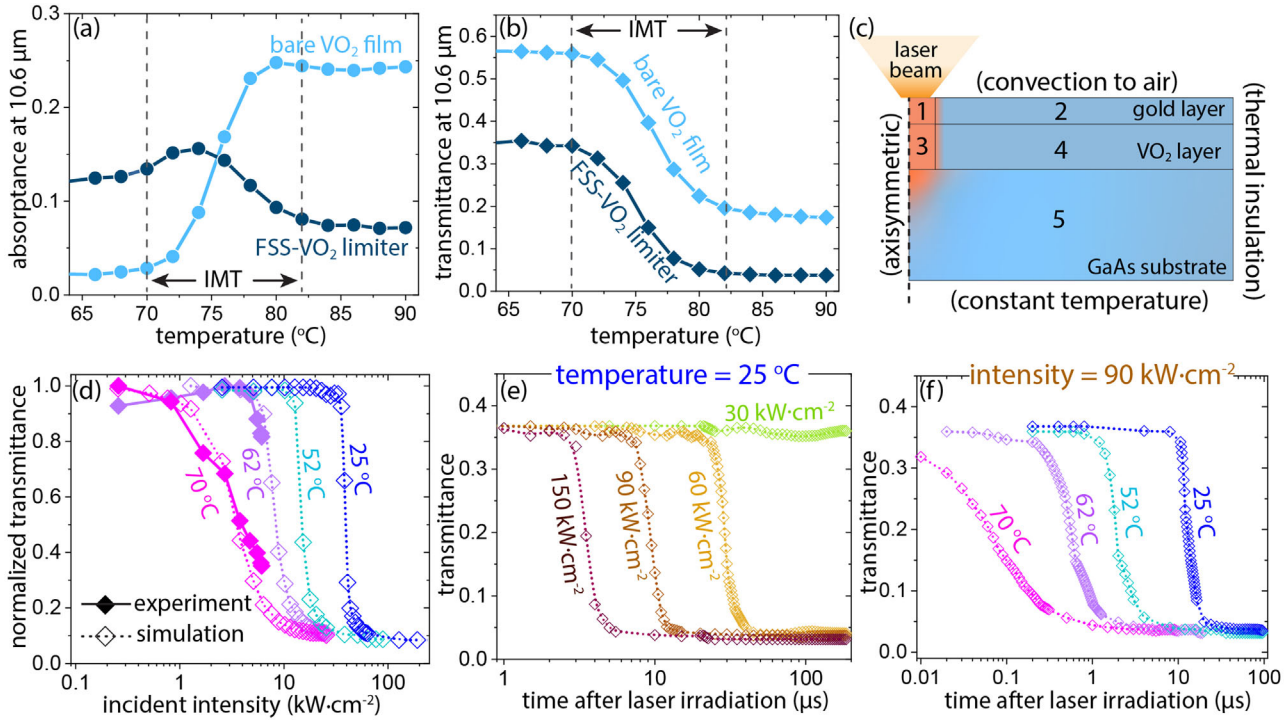

Figure 4. Measured temperature-dependent a) absorptance and b) transmittance of both the bare VO_2 film and FSS-VO_2 limiter at $\lambda = 10.6$ μm. c) Geometry for photothermal simulations. The axisymmetric model consists of five domains of which 1 and 3 experience absorption due to the laser irradiation. d) Comparison between the simulated power-dependent transmittance based on the model in (c) and the experimental results in Figure 3c. e) Simulated time-resolved transmittance of the FSS-VO_2 limiter for incident intensities of 30, 60, 90, and 150 kW·cm^{-2}, when it is used as a passive device at 25 °C (i.e., without thermal bias). f) Simulated time-resolved transmittance of the FSS limiter for an incident intensity of 90 kW·cm^{-2}, when it is thermally biased at 25, 52, 62, and 70 °C.

opto-thermal simulation loop iterates until the temperature distribution stabilizes. Finally, we can convert the stabilized temperature distribution to transmittance according to Figure 4b. Based on convergence tests, we chose to use a step of 0.1 μs and a total simulation time of 500 μs (see details in Supporting Information S6).

Both the limiting threshold and the trend of reduction of transmittance with incident intensity are well captured in our simulations (Figure 4d), and agree very well with experimental data extracted from Figure 3c (see Supporting Information S4 for more details). Our simulations also confirmed that our experiments never fully reached the complete metal phase of the VO_2 due to the limited incident intensities (<6.5 kW·cm^{-2}). One can expect that once the IMT is completed, the output power will start to increase again with a very slow rate of ≈0.03 (i.e., the limiting-state transmittance) as the input power increases.

Our opto-thermal modeling can be used to estimate the limiting threshold for a passive device (i.e., without thermal biasing) and the response time of the device (i.e., the time needed to reduce transmittance by e^{-1}), which are important parameters for practical use. In our experiments, we did not demonstrate nonlinear transmittance without thermal bias due to the limited output power of our laser. Our simulations predicted that, at room temperature (25 °C), the limiting state will be triggered when the incident intensity reaches ≈40 kW·cm^{-2} (blue curve in Figure 4d). The response time was obtained by converting the temperature distribution at each time step to the transmittance. For the limiter working at 25 °C, the time evolution of the transmittance for different incident intensities is shown in Figure 4e. For example, given an incident intensity of 60 kW·cm^{-2}, the response time is ≈30 μs. The response time reduces to <4 μs when the incident intensity is greater than 150 kW·cm^{-2}. Note that the recovery time (from the limiting state to the open state, after the light turns off) maybe much longer than the limiting response time, but the recovery time is not particularly important for limiting applications. Similarly, for a given incident intensity (e.g., 90 kW·cm^{-2}) one can expect to have a faster response at a higher bias temperature (Figure 4f). However, passive devices are more practical. In the absence of thermal bias, the limiting threshold can be engineered via a reduction of the IMT temperature using doping[42] or defect engineering.[43] On the other hand, to increase the limiting threshold, one can reduce the open-state absorption by redesigning the FSS.[44]

Compared to conventional optical limiters that are based on nonlinear absorption such as reverse saturable absorption or multi-photon absorption, our reflective limiter has two advantages: low triggering threshold and high damage threshold. Due to the ultrahigh photothermally induced nonlinearity of VO_2, our limiter can feature limiting behavior for incident intensities of ≈1–100 kW·cm^{-2} (as shown in Figures 3 and 4), while nonlinear-absorption materials usually need the intensity to be higher than 10^5 kW·cm^{-2} to trigger the nonlinearity.[6,10] We note that Howes et al.[45] recently demonstrated an optical limiter by incorporating VO_2 with dielectric metasurfaces, which also has a similar low

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triggering threshold as that of our design. Moreover, our limiter is designed to be highly reflective in the limiting state to avoid the damage (such as melting or delamination) that can be caused by the absorption of the intense light. On the other hand, conventional nonlinear-absorption limiters have their own advantages such as low fabrication cost and broad bandwidth since they usually do not require optical-resonance structures.

For practical applications, it is also important to consider the effect of the hysteresis in VO_2.[46] For many applications that may be enabled by VO_2, like modulators or switches, it is necessary that the device can be turned on and off quickly. On the other hand, in an optical limiter, the return to the open state when no longer illuminated by high-power light is ideal but not absolutely necessary. Nevertheless, hysteresis can be overcome in our optical limiters if the device is biased outside of the hysteresis loop (e.g., temperatures smaller than 52 °C according to Figure S3c in Supporting Information). Our simulations predict that, when biased at 52 °C, the nonlinear transmittance can be achieved when the incident intensity surpasses 15 $kW \cdot cm^{-2}$ (the cyan curve in Figure 4d). Alternatively, one can also consider exploring phase transitions with minimal hysteresis (e.g., the IMT in samarium nickelate[47]).

Finally, we note that in our experimental demonstration, the IMT in VO_2 was photothermally triggered by a continuous-wave laser. Therefore, the device speed is limited to approximately microseconds. However, the IMT in VO_2 can actually be triggered nonthermally on time scales as short as tens of femtoseconds using high-intensity optical pulses,[48,49] and therefore we anticipate that VO_2-based limiters can also limit very-high-intensity pulses at these time scales. The optimization of VO_2-based limiters for ultrafast operation may require a redesign, optimizing for higher electric-field concentration within the VO_2 layer to more-easily trigger the IMT.[50]

5. Conclusion

Reflective optical limiters, which are transmissive at low input powers but reflecting at high powers, are a promising technology for sensor protection, because they can avoid damage due to light absorption in the limiter itself. Here, we explored the use of resonant transmission through metallic frequency-selective surfaces (FSSs) as the basis for reflective limiters, using the IMT in vanadium dioxide (VO_2) to make the resonant transmittance sensitive to incident intensity. The IMT in VO_2 can be a source of giant photothermal nonlinearity, which enabled the use of low-quality-factor FSS resonances in our design, resulting in open-state transmittance with large angular and temporal bandwidth. Our prototype reflective limiter designed for a wavelength of 10.6 µm has high open-state transmittance (≈0.7), low limiting-state transmittance (<0.01) and absorptance (≈0.06), broad working bandwidth (FWHM > 2 µm), and functions for all polarizations up to an incident angle of ≈50°.

6. Experimental Section

VO_2 was deposited onto double-side-polished undoped GaAs (001) wafers via magnetron sputtering from a V_2O_5 target, with radio-frequency power of 100 W. During deposition, the chamber pressure was maintained at 5 mTorr with an Ar/O_2 gas mixture at a flow rate of 49.85/0.15 sccm. The substrate was heated to 700 °C to form the VO_2 phase.

We built a 50-nm gold FSS on top of the as-grown VO_2 via the following steps. First, a ≈250-nm PMMA (495 PMMA A4) was spin-coated onto the VO_2 film and the pattern was written using an e-beam lithography system (Elionix GS-100). After development in MIBK/IPA (volume ratio of 1/3), an array of cross-shape PMMA blocks were left on top of the VO_2 film. Then, 50 nm of gold was evaporated, and the sample was soaked in an acetone bath for a few minutes to lift-off the gold-PMMA blocks, leaving an array of cross-slit apertures within the gold film. Note that a 60-s sonication was necessary during the lift-off process.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

frequency-selective surfaces, optical limiter, optical nonlinearity, phase-change materials, vanadium dioxide

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[1] Y. Izawa, N. Miyanaga, J. Kawanaka, K. Yamakawa, *J. Opt. Soc. Korea* **2008**, *12*, 178.
[2] S. D. Jackson, *Nat. Photonics* **2012**, *6*, 423.
[3] X. Michalet, O. H. W. Siegmund, J. V. Vallerga, P. Jelinsky, J. E. Millaud, S. Weiss, *J. Modern Optics* **2007**, *54*, 239.
[4] J. Miao, F. Zhang, *J. Mater. Chem. C* **2019**, *7*, 1741.
[5] A. E. Siegman, *Appl. Opt.* **1962**, *1*, 127.
[6] L. W. Tutt, T. F. Boggess, *Prog. Quantum Electron.* **1993**, *17*, 299.
[7] M. J. Miller, A. G. Mott, B. P. Ketchel, in *Nonlinear Optical Liquids for Power Limiting and Imaging*, **1998**, *3472*, pp. 24–29.
[8] C. W. Spangler, *J. Mater. Chem.* **1999**, *9*, 2013.
[9] D. Dini, M. J. F. Calvete, M. Hanack, *Chem. Rev.* **2016**, *116*, 13043.
[10] Y. P. Sun, J. E. Riggs, *Int. Rev. Phys. Chem.* **1999**, *18*, 43.

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[11] D. J. Hagan, S. Guha, E. W. Van Stryland, M. J. Soileau, Y. Y. Wu, *Opt. Lett.* **1988**, *13*, 315.
[12] N. Venkatram, D. N. Rao, M. A. Akundi, *Opt. Express* **2005**, *13*, 867.
[13] L. Polavarapu, Q. H. Xu, M. S. Dhoni, W. Ji, *Appl. Phys. Lett.* **2008**, *92*, 263110.
[14] H. Qian, S. Li, Y. Li, C. F. Chen, W. Chen, S. E. Bopp, Y. U. Lee, W. Xiong, Z. Liu, *Sci. Adv.* **2020**, *6*, eaay3456.
[15] Z. Liu, B. Zhang, N. Dong, J. Wang, Y. Chen, *J. Mater. Chem. C* **2020**, *8*, 10197.
[16] L. Wang, *Liq. Cryst.* **2016**, *43*, 2062.
[17] Y. Chen, T. Bai, N. Dong, F. Fan, S. Zhang, X. Zhuang, J. Sun, B. Zhang, X. Zhang, J. Wang, W. J. Blau, *Prog. Mater. Sci.* **2016**, *84*, 118.
[18] E. W. Van Stryland, Y. Y. Wu, D. J. Hagan, M. J. Soileau, K. Mansour, *J. Opt. Soc. Am. B* **1988**, *5*, 1980.
[19] W. Li, Y. Wang, G. V. Naik, In *CLEO: QELS Fundamental Science, Optical Society of America* **2020**, pp. FF3Q-3.
[20] G. Liang, L. Tao, Y. H. Tsang, L. Zeng, X. Liu, J. Li, J. Qu, Q. Wen, *J. Mater. Chem. C* **2019**, *7*, 495.
[21] R. C. Hollins, *Curr. Opin. Solid State Mater. Sci.* **1999**, *4*, 189.
[22] J. H. Vella, J. H. Goldsmith, A. T. Browning, N. I. Limberopoulos, I. Vitebskiy, E. Makri, T. Kottos, *Phys. Rev. Appl.* **2016**, *5*, 064010.
[23] A. Sarangan, J. Duran, V. Vasilyev, N. Limberopoulos, I. Vitebskiy, I. Anisimov, *IEEE Photonics J.* **2018**, *10*, 1.
[24] C. Wan, E. H. Horak, J. King, J. Salman, Z. Zhang, Y. Zhou, P. Roney, B. Gundlach, S. Ramanathan, R. H. Goldsmith, M. A. Kats, *ACS Photonics* **2018**, *5*, 2688.
[25] W. Wang, Y. Luo, D. Zhang, F. Luo, *Appl. Opt.* **2006**, *45*, 3378.
[26] C. Wan, Z. Zhang, D. Woolf, C. M. Hessel, J. Rensberg, J. M. Hensley, Y. Xiao, A. Shahsafi, J. Salman, S. Richter, Y. Sun, M. M. Qazilbash, R. Schmidt-Grund, C. Ronning, S. Ramanathan, M. A. Kats, *Ann. Phys.* **2019**, *531*, 1900188.
[27] Z. Yang, C. Ko, S. Ramanathan, *Annu. Rev. Mater. Res.* **2011**, *41*, 337.
[28] C. Kübler, H. Ehrke, R. Huber, R. Lopez, A. Halabica, R. F. Haglund, A. Leitenstorfer, *Phys. Rev. Lett.* **2007**, *99*, 116401.
[29] C. Wan, E. H. Horak, J. King, J. Salman, Z. Zhang, Y. Zhou, P. Roney, B. Gundlach, S. Ramanathan, R. H. Goldsmith, M. A. Kats, **2018**, *5*, 2688.
[30] J. Leuthold, C. Koos, W. Freude, *Nat. Photonics* **2010**, *4*, 535.
[31] R. W. Boyd, *Nonlinear Optics*, Elsevier Inc., Amsterdam **2008**.
[32] J.-F. Xu, R. Czerw, S. Webster, D. L. Carroll, J. Ballato, R. Nesper, *Appl. Phys. Lett.* **2002**, *81*, 1711.
[33] O. B. Danilov, V. A. Klimov, O. P. Mikheeva, A. I. Sidorov, S. A. Tul'skii, E. B. Shadrin, I. L. Yachnev, *Tech. Phys.* **2003**, *48*, 73.
[34] P. B. Johnson, R. W. Christy, *Phys. Rev. B* **1972**, *6*, 4370.
[35] T. Skauli, P. S. Kuo, K. L. Vodopyanov, T. J. Pinguet, O. Levi, L. A. Eyres, J. S. Harris, M. M. Fejer, B. Gerard, L. Becouarn, E. Lallier, *J. Appl. Phys.* **2003**, *94*, 6447.
[36] J. Rensberg, "Defect engineering of phase transition materials to create optical metasurfaces", *Doctoral dissertation*, **2018**, *Appendix Section 4.*.
[37] H. Zhang, Z. Wu, Q. He, Y. Jiang, *Appl. Surf. Sci.* **2013**, *277*, 218.
[38] S. Lee, K. Hippalgaonkar, F. Yang, J. Hong, C. Ko, J. Suh, K. Liu, K. Wang, J. J. Urban, X. Zhang, C. Dames, S. A. Hartnoll, O. Delaire, J. Wu, *Science* **2017**, *355*, 371.
[39] E. Freeman, A. Kar, N. Shukla, R. Misra, D. Schlom, V. Gopalan, K. Rabe, S. Datta, **2009**, *56*, 10.
[40] J. C. Yannopoulos, *In The Extractive Metallurgy of Gold*, Springer US, Boston, MA, **1991**, p. 11.
[41] https://www.janis.com/docs/default-source/default-document-library/gallium-arsenide-gaas-transmission-curve-datasheet.pdf?sfvrsn=a610d3d9_2
[42] T. D. Manning, I. P. Parkin, M. E. Pemble, D. Sheel, D. Vernardou, *Chem. Mater.* **2004**, *16*, 744.
[43] J. Rensberg, S. Zhang, Y. Zhou, A. S. McLeod, C. Schwarz, M. Goldflam, M. Liu, J. Kerbusch, R. Nawrodt, S. Ramanathan, D. N. Basov, F. Capasso, C. Ronning, M. A. Kats, *Nano Lett.* **2016**, *16*, 1050.
[44] F. J. Garcia-Vidal, L. Martin-Moreno, T. W. Ebbesen, L. Kuipers, *Rev. Mod. Phys.* **2010**, *82*, 729.
[45] A. Howes, Z. Zhu, D. Curie, J. R. Avila, V. D. Wheeler, R. F. Haglund, J. G. Valentine, *Nano Lett.* **2020**, *20*, 4638.
[46] M. Gurvitch, S. Luryi, A. Polyakov, A. Shabalov, *J. Appl. Phys.* **2009**, *106*, 104504.
[47] A. Shahsafi, P. Roney, Y. Zhou, Z. Zhang, Y. Xiao, C. Wan, R. Wambold, J. Salman, Z. Yu, J. Li, J. T. Sadowski, R. Comin, S. Ramanathan, M. A. Kats, *Proc. Natl. Acad. Sci. U. S. A.* **2019**, *116*, 26402.
[48] M. F. Jager, C. Ott, P. M. Kraus, C. J. Kaplan, W. Pouse, R. E. Marvel, R. F. Haglund, D. M. Neumark, S. R. Leone, *Proc. Natl. Acad. Sci. U. S. A.* **2017**, *114*, 9558.
[49] T. Kang, Z. Ma, J. Qin, Z. Peng, W. Yang, T. Huang, S. Xian, S. Xia, W. Yan, Y. Yang, Z. Sheng, J. Shen, C. Li, L. Deng, L. Bi, *Nanophotonics* **2020**, *10*, 909.
[50] M. Liu, H. Y. Hwang, H. Tao, A. C. Strikwerda, K. Fan, G. R. Keiser, A. J. Sternbach, K. G. West, S. Kittiwatanakul, J. Lu, S. A. Wolf, F. G. Omenetto, X. Zhang, K. A. Nelson, R. D. Averitt, *Nature* **2012**, *487*, 345.

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